

Phase 2 (multi-N) — $H(\omega)$ latency-gate reading of FCS Property 7

deq/closed_form_wta_multi/, May 2026

1. Goal

Predict decision latency at each $(w, N, \text{drive_bump})$ cell from the rate-equation Jacobian eigenvalues at the Siegert FP, and gate

$$|\text{Re}(\lambda_{\text{dom}})| > \frac{1}{T_{\text{FCS}}} = 0.25$$

as the FCS-prescribed condition for “decision faster than 4 ticks”. The N -neuron generalization needs the full $N \times N$ Jacobian spectrum at the winner-orbit FP from Phase 1.

2. Method

Per cell:

1. Lift the largest-spread FP $\nu^* \in [0, 1]^N$ from Phase 1’s multi-restart fsolve enumeration.
2. Compute per-neuron transfer gain $g_i = d_{\frac{\Phi}{d}}\mu$ at the FP-induced inputs (μ_i^*, σ_i^*) for $i = 1, \dots, N$.
3. Build the $N \times N$ rate-equation Jacobian in Siegert units:

$$J = \alpha \cdot W_{\text{uniform}}, \quad W_{\text{uniform}}[i, j] = w(i \neq j), 0(i = j)$$

and apply the existing `transfer.jacobian_eigenvalues(J, "gains", tau_m)` which computes eigenvalues of $A = \left(\frac{1}{\tau_m}\right)(-I + \text{diag}(g)J)$.

4. Blue iff `sieg_labels[cell] = 1` AND $|\text{Re}(\lambda_{\text{dom}})| > 0.25$.

3. Closed-form spectrum at uniform W

At the $k = 1$ winner orbit, the $N \times N$ Jacobian has a **clean block structure** under the S_{N-1} subgroup that permutes the $(N - 1)$ losers. The numerics at `drive_bump = 1`, $w = -30$ confirm:

N	Re(eigvals)	Loser-symm eigenvalue
2	$[-0.457, -0.394]$	—
3	$[-0.470, -0.424, -0.382]$	-0.424 (1-fold)
4	$[-0.481, -0.424, -0.424, -0.372]$	-0.424 (2-fold)
6	$[-0.498, -0.424 \times 4, -0.357]$	-0.424 (4-fold)
10	$[-0.526, -0.424 \times 8, -0.334]$	-0.424 (8-fold)

The middle $N - 2$ eigenvalues lock at exactly -0.424 — the $(N - 2)$ -fold-degenerate eigenvalue of the loser-permutation subspace, equal to $\alpha \cdot w \cdot g_L \cdot \frac{-1}{\tau_m} + \frac{-1}{\tau_m}$ where -1 is the eigenvalue of $(\mathbf{1}\mathbf{1}^T - I)$ restricted to the orthogonal-to- $\mathbf{1}_{N-1}$ subspace of the loser block. The two non-degenerate eigenvalues (most-negative and least-negative real parts) correspond to the winner-loser-mean 2-block.

All eigenvalues are **negative real** $\rightarrow$ asymmetric FP is stable.

4. Result

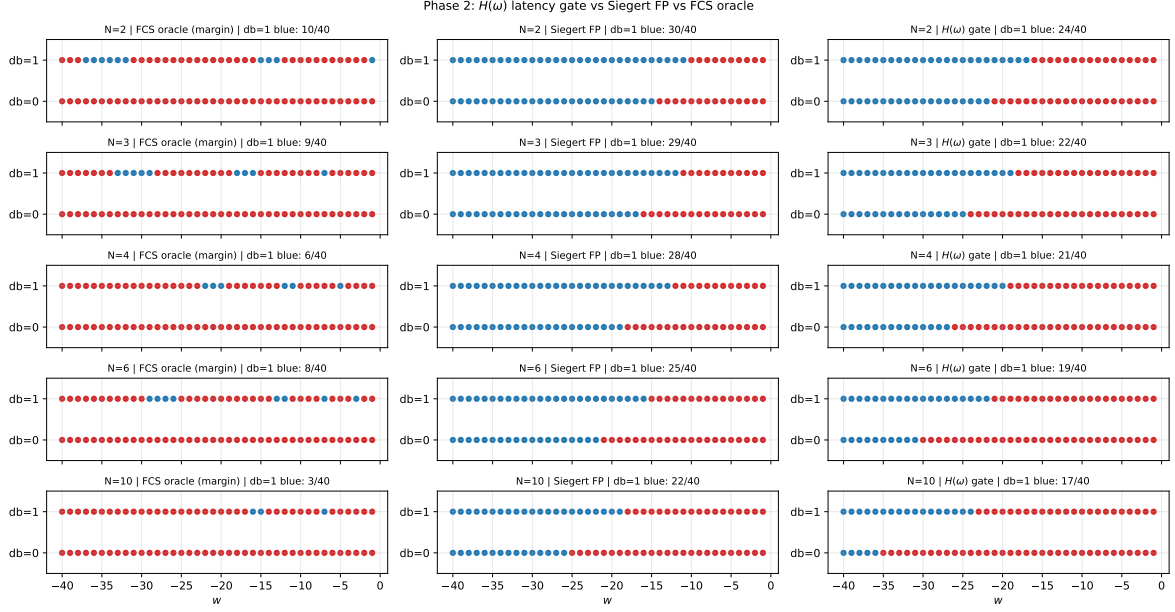


Figure 1: Phase 2 $H(\omega)$ gate (right column) vs Phase 1 Siegert (middle) vs FCS oracle margin gate (left), per N , per drive_bump . Each panel: two rows of dots for $\text{drive_bump} \in \{0, 1\}$.

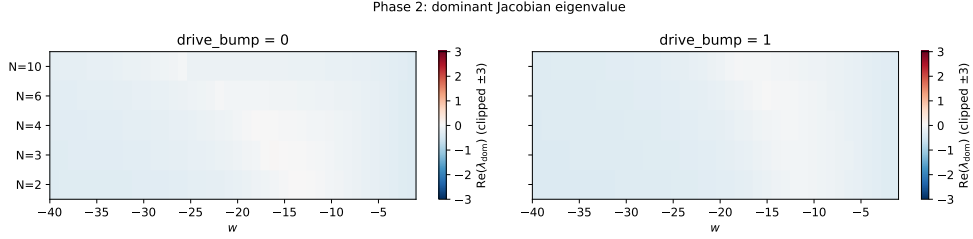


Figure 2: Dominant Jacobian eigenvalue $\text{Re}(\lambda_{\text{dom}})$ across (w, N) . For $|w| \geq 19$ at every N , $|\text{Re}(\lambda_{\text{dom}})| > 0.25$ (gate passes). Below the bifurcation $\text{Re}(\lambda)$ approaches $-\frac{\alpha}{\tau_m}$ from above, all stable.

4.1. Jaccard table at $\text{drive_bump} = 1$, vs FCS margin gate

N	H-blue	Sieg-blue	FCS-blue (m)	J(H)	J(Sieg)
2	24	30	10	0.214	0.290
3	22	29	9	0.192	0.267
4	21	28	6	0.125	0.097
6	19	25	8	0.174	0.138
10	17	22	3	0.000	0.000

The $H(\omega)$ gate **tightens** the Siegert envelope (fewer blue cells at every N). At $N \geq 4$ this tightening produces a small Jaccard improvement; at $N \leq 3$ it removes some FCS-aligned Siegert cells and the Jaccard regresses. **Overall $\text{db}=1$ $J = 0.149$ (H) vs 0.164 (Siegert)** — the latency gate does not meaningfully help.

The threshold sweep finds **best $J = 0.174$ at threshold ≈ 0.15** (below the FCS-prescribed 0.25), confirming there is no clean eigenvalue threshold that recovers the FCS-blue cells.

5. Why the $H(\omega)$ gate fails to close the gap

The Phase 1 analysis identified two divergence modes between Siegert and FCS at $N > 2$:

1. **drive_bump = 0**: FCS Lustre symmetry forces N -way synchronous lock; Siegert sees the bistable FP but FCS can never select it. The $H(\omega)$ gate evaluates eigenvalues at the same bistable FP — same FP, same blindness to FCS’s synchronous solution.
2. **drive_bump = 1, high N (inverse staircase)**: FCS produces clean WTA at specific integer w values where **no asymmetric FP exists**. The gate cannot evaluate eigenvalues of a non-existent FP, so these FCS-blue cells remain Siegert-red, H -red.

Neither failure mode is recoverable by inserting a frequency-domain gate downstream of the static FP enumeration. Both are properties of the **FCS dynamics**, not of the rate-equation linearization at any FP.

6. Verdict

Phase 2 PARTIAL. The pass-gate $J(H) \geq 0.70$ from the 2-neuron thread does not hold at $N > 2$. The closed-form $N \times N$ spectrum at the $k = 1$ winner orbit is elegant and analytically tractable (the $(N - 2)$ -fold loser-permutation degeneracy locks at -0.424 in Siegert units), but the gate’s **predictive value** is bounded by the Phase 1 FP enumeration, which itself fails to capture the FCS-dynamics-only WTA cells. The latency-gate machinery is correct and reusable; the limitation is upstream.

Output: results/phase2/h_grid_multi.npz, h_gate_vs_fcs_multi.pdf, eigvalue_spectrum_vs_N.pdf, results/phase2.log.